

Reduction of Singularities in Poisson Geometry via Weighted Blowups

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X - complex manifold, $\mathcal{O}_X$ - sheaf of holomorphic functions.

Definition

A **Poisson bracket** on X is a bilinear morphism

$$\{\cdot, \cdot\}: \mathcal{O}_X \times \mathcal{O}_X \rightarrow \mathcal{O}_X$$

satisfying:

- ① (Skew-symmetry) $\{f, g\} = -\{g, f\}$,
 - ② (Leibniz rule) $\{fg, h\} = f\{g, h\} + g\{f, h\}$,
 - ③ (Jacobi identity) $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.
- (X, π) is a **Poisson manifold**.

Equivalently: **Poisson bivector-field**

$$\pi = \sum_{1 \leq i < j \leq n} \{x^i, x^j\} \frac{\partial}{\partial x^i} \wedge \frac{\partial}{\partial x^j} \quad \text{with} \quad [\pi, \pi] = 0.$$

(X, π) - Poisson manifold

Definition

A subvariety $Y \subset X$ is called a **Poisson (sub)variety** if $\mathcal{I}_Y \subset \mathcal{O}_X$ is a Poisson ideal:

$$\{\mathcal{I}_Y, \mathcal{O}_X\} \subset \mathcal{I}_Y.$$

The Poisson bracket on $\mathcal{O}_X$ descends to a bracket on $\mathcal{O}_Y = \mathcal{O}_X/\mathcal{I}_Y$:

$$\{\cdot, \cdot\}: \frac{\mathcal{O}_X}{\mathcal{I}_Y} \times \frac{\mathcal{O}_X}{\mathcal{I}_Y} \longrightarrow \frac{\mathcal{O}_X}{\mathcal{I}_Y}$$

- Any symplectic manifold X ; e.g. a cotangent bundle $X = T^*Q$:

$$\pi = \frac{\partial}{\partial q^i} \wedge \frac{\partial}{\partial p_i}.$$

Only Poisson ideal is $\mathcal{I}_Y = 0$, hence only Poisson subvariety is $Y = X$.

- The dual $\mathfrak{g}^*$ of any Lie algebra $\mathfrak{g}$:

$$[e_i, e_j]_{\mathfrak{g}} = c_{ij}^k e_k \iff \{x_i, x_j\}_{\mathfrak{g}^*} = c_{ij}^k x_k \iff \pi = c_{ij}^k x_k \frac{\partial}{\partial x_i} \wedge \frac{\partial}{\partial x_j}.$$

Any Lie algebra ideal $\mathfrak{h} \triangleleft \mathfrak{g}$ defines a Poisson ideal $\mathfrak{h} \triangleleft C^\infty(\mathfrak{g}^*)$, hence $Y = \mathbf{V}(\mathfrak{h}) = \mathfrak{h}^\circ$ is a Poisson subvariety.

- $X = \mathbb{C}^3$. Any holomorphic function f defines a Poisson bivector field:

$$\pi = \iota_{df} \partial_1 \wedge \partial_2 \wedge \partial_3 = \overbrace{[\partial_1 \wedge \partial_2 \wedge \partial_3, f]}^{\text{Schoutten bracket}}$$

so that $Y = \mathbf{V}(f)$ is a Poisson subvariety. Moreover, every level set $\mathbf{V}(f - c)$ is also a Poisson subvariety.

Example: $\mathfrak{sl}_2^*(\mathbb{C}) = \mathfrak{so}_3^*(\mathbb{C})$

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$X = \mathbb{C}^3$ with $\pi = x\partial_y \wedge \partial_z + y\partial_z \wedge \partial_x - z\partial_x \wedge \partial_y = \frac{1}{2}[\partial_x \wedge \partial_y \wedge \partial_x, x^2 + y^2 - z^2]$.



Singular Poisson subvariety (red):

$$Y = \mathbf{V}(\overbrace{x^2 + y^2 - z^2}^f).$$

Indeed:

$$\begin{aligned}\{x, f\} &= \{x, x^2 + y^2 - z^2\} \\ &= 2x\{x, x\} + 2y\{x, y\} - 2z\{x, z\} \\ &= -2yz + 2zy \\ &= 0.\end{aligned}$$

Similarly $\{y, f\} = \{z, f\} = 0$,

$$\implies \{\mathcal{I}_Y, \mathcal{O}_X\} \subset \mathcal{I}_Y.$$

(X, Y, π) - Poisson triple (meaning $Y \subset (X, \pi)$ Poisson subvariety of a Poisson manifold)

Definition

A **reduction of singularities** of (X, Y, π) is a Poisson map

$$\sigma: (X', Y', \pi') \longrightarrow (X, Y, \pi)$$

satisfying:

- 1 σ is proper, surjective,
- 2 (X', Y') have strictly “better” singularities: $\text{inv}(X', Y') < \text{inv}(X, Y)$,
- 3 Isomorphism away from Y_{sing} .

It is a **Poisson resolution** when (X', Y') are smooth.

Q: (When) does Poisson resolution exist?

Dream: Use an existing resolution algorithm. E.g. in the absence of a Poisson structure:

- ① Hironaka [Hir64]: resolution via **blowups**.
- ② Abramovich, Temkin, Włodarczyk [ATW24]: functorial embedded resolution via **weighted blowups**.

$$(Y^{(n)} \subset X^{(n)}) \xrightarrow{\phi^{(n)}} \dots \xrightarrow{\phi^{(2)}} (Y^{(1)} \subset X^{(1)}) \xrightarrow{\phi^{(1)}} (Y \subset X).$$

Reality: Multivectors (e.g. Poisson structures) don't always pullback.

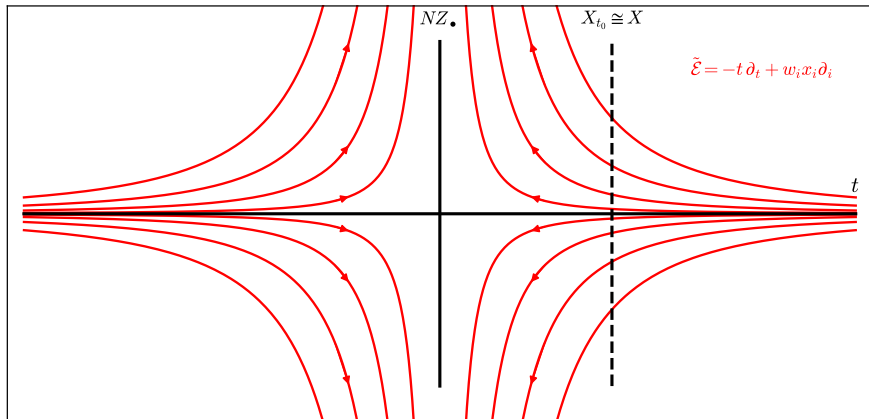
Q': How far can Poisson reduction of singularities go, using (weighted) blowups?

Weighted Blowups of Poisson Structures

There is a **lifting criterion** for doing weighted blowups of Poisson structures.

Lemma (Lapointe-Matviichuk-Pym-Z.)

A Poisson structure π lifts to a weighted blowup $\sigma: \text{Bl}_{Z_\bullet}(X, Y) \rightarrow (X, Y)$ with lift tangent to $E = \sigma^{-1}Z_\bullet$ if and only if $\text{ord}_{Z_\bullet} \pi \geq 0$.



$X = \mathbb{C}^3$, $Y = \mathbf{V}(x^2 + y^2 - z^2)$ with $\pi = x\partial_y \wedge \partial_z + y\partial_z \wedge \partial_x - z\partial_x \wedge \partial_y$.



Any weighted center

$$Z_{\bullet} = \mathbf{V}(x^{1/w_x}, y^{1/w_y}, z^{1/w_z}) \subset Y_{\text{sing}} = \{0\}$$

must have $w_x > 0$, $w_y > 0$, $w_z > 0$.

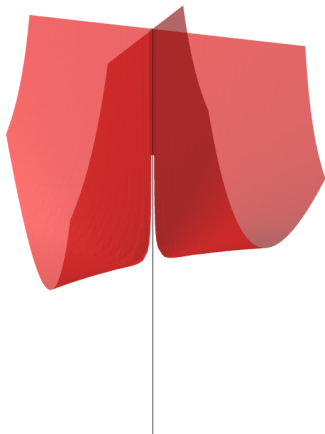
In this case:

$$\begin{aligned} \text{ord}_{Z_{\bullet}} \pi = \min \{ & w_x - w_y - w_z, \\ & w_y - w_z - w_x, \\ & w_z - w_x - w_y \} < 0. \end{aligned}$$

$\implies$ No way to resolve (X, Y, π) with weighted blowups.

Example: Whitney Umbrella

$$X = \mathbb{C}^3, Y = \mathbf{V}(x^2 - y^2z) \text{ with } \pi = x\partial_y \wedge \partial_z - zy\partial_z \wedge \partial_x - \frac{1}{2}y^2\partial_x \wedge \partial_y.$$



Any weighted center

$$Z_{\bullet} = \mathbf{V}(x^{1/w_x}, y^{1/w_y}, z^{1/w_z}) \subset Y_{\text{sing}} = \mathbf{V}(x, y)$$

must have $w_x > 0$, $w_y > 0$, $w_z \geq 0$. In this case:

$$\begin{aligned} \text{ord}_{Z_{\bullet}} \pi = \min \{ & w_x - w_y - w_z, \\ & w_y - w_x, \\ & w_y - w_x \} \geq 0. \end{aligned}$$

$$\iff w_z = 0, w_x = w_y.$$

$\implies$ Can't blowup with $Z_{\bullet} = \mathbf{V}(x^2, y^3, z^3)$. $\implies$
Can blowup with $Z_{\bullet} = \mathbf{V}(x^2, y^2)$; this resolves (X, Y, π) .

Theorem (Lapointe-Matviichuk-Pym-Z.)

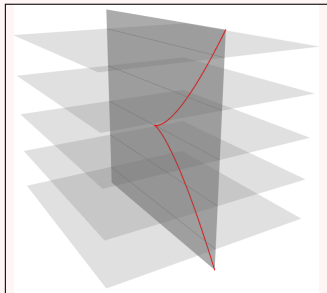
(X, Y, π) with $\dim X \leq 3$. There exists a sequence of weighted blowups

$$(X', Y', \pi') \longrightarrow \cdots \longrightarrow (X, Y, \pi)$$

such that (X', Y', π') is either smooth, or has one of the following singularities,

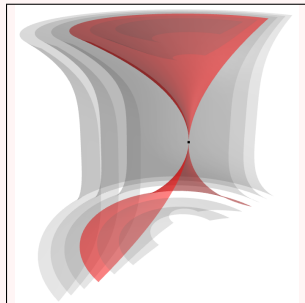
Non-nilpotent singularity:

$$Y' = \{x = g(y, z) = 0\}, \pi' = x\partial_x \wedge \partial_y$$



du Val surface singularity:

$$Y' = \{f = 0\}, \pi' = g[\partial_x \wedge \partial_y \wedge \partial_z, f]$$

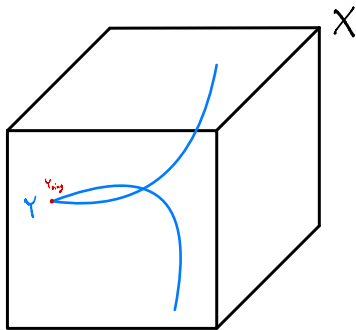


Y_{sing} is isolated and

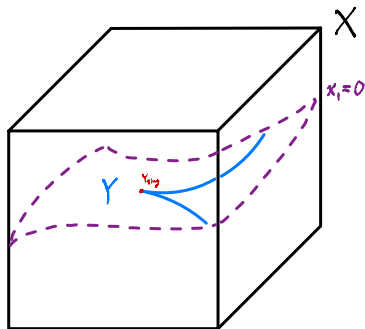
$$\pi = \phi_1 \cdot \sigma_1 + \cdots + \phi_r \cdot \sigma_r$$

where $Y = \mathbf{V}(\phi_1, \dots, \phi_r)$ and $\sigma_1, \dots, \sigma_r \in \mathfrak{X}_X^2$.

Space Curve

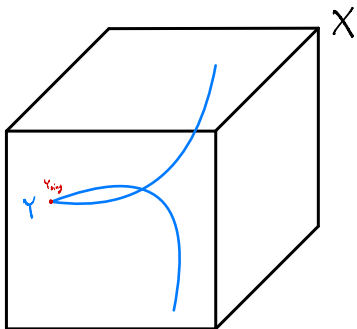


Plane Curve



Space Curve

$$Y = \mathbf{V}(\phi_1, \dots, \phi_r)$$



- ① $\phi_1, \dots, \phi_r$ must be at least quadratic.



A straightforward computation shows that

$$\text{ord}_{Z_\bullet} \pi \geq 0$$

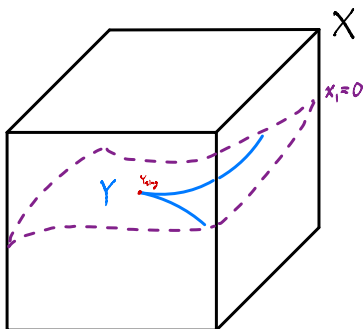
when $Z_\bullet$ is the weighting of [ATW24].

- ② The weighted blowup of [ATW24] always improves the singularities.

$\implies \pi$ lifts to the weighted blowup.

Plane Curve

$$Y = \mathbf{V}(\phi_1, \dots, \phi_r)$$



Y is (locally) trapped in a coordinate hypersurface $\{x_1 = 0\}$ where $x_1 = \phi_1$.

Case 1: If π is at least quadratic, then π is inflatable under the ordinary blowup.^a

Case 2: If π has a linear term π_{lin} , then either
 $\pi_{\text{lin}} = x_1 \partial_1 \wedge \partial_2$ or $\pi_{\text{lin}} = x_1 \partial_2 \wedge \partial_3$.

By deformation theory for Poisson structures:

$$\begin{array}{ll} \pi = x_1 \partial_1 \wedge \partial_2 & \text{or} \quad \pi = x_1 \partial_2 \wedge \partial_3 + (\text{h.w.t}) \\ \text{ord}_{Z_\bullet} \pi = -w_2 & \text{ord}_{Z_\bullet} \pi = w_1 - w_2 - w_3 \\ \text{non-inflatable} & \text{inflatable by [ATW24]} \end{array}$$

^aA result of [Abhyankar-1983] guarantees that the singularity invariant decreases.

A normal form result of [Frankel 2013] says:

$$\pi = g[\partial_1 \wedge \partial_2 \wedge \partial_3, f] + f\eta.$$

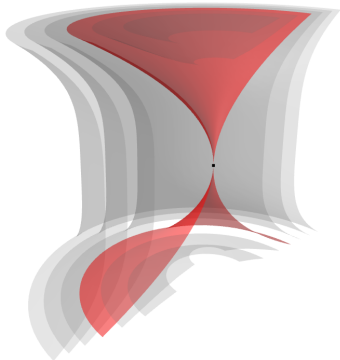
Let $Z_\bullet$ be the ATW center. If $\text{ord}_{Z_\bullet} \pi < 0$, then:

$$\text{inv}(X, Y) = (a_1, a_2, a_3) < (2, 3, 6).$$

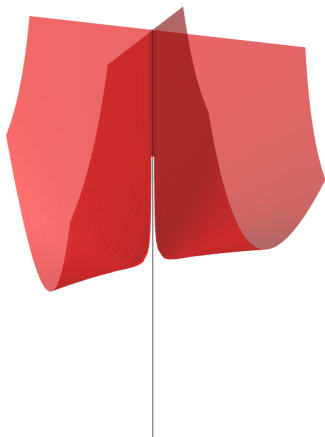
These are locally $Y = \mathbf{V}(f)$ where

Name	f
$A_n, n \geq 1$	$x^2 + y^2 + z^{n+1}$
$D_n, n \geq 4$	$x^2 + y^2 z + z^{n-1}$
E_6	$x^2 + y^3 + z^4$
E_7	$x^2 + y^3 + yz^3$
E_8	$x^2 + y^3 + z^5$

Isolated Surface Singularity



Non-Isolated Singularity



Only other case to consider is the Whitney umbrella (complicated proof):

$$Y = \mathbf{V}(x^2 - y^2z)$$

$$\pi = x\partial_y \wedge \partial_z - yz\partial_z \wedge \partial_x - \frac{1}{2}y^2\partial_x \wedge \partial_y$$

From example: $Z_{\bullet} = \mathbf{V}(x^2, y^2)$ resolves (X, Y, π) .

- [ATW24] Dan Abramovich, Michael Temkin, and Jarosław Włodarczyk, *Functorial embedded resolution via weighted blowings up*, Algebra & Number Theory **18** (2024), no. 8, 1557–1587.
- [LMPZ26] Simon Lapointe, Mykola Matviichuk, Brent Pym, and Boris Zupancic, *Weighted blowups and 3d Poisson desingularizations*, April 2026.